



SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005

ADVANCED TECHNOLOGY SERVICES – HMA

1800 Honda Drive

Lincoln, AL 35096

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CALIBRATION

Valid to: March 31, 2017

Certificate Number: 1592.04

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations¹:

I. Dimensional

Parameter/Equipment	Range	CMC ² (±)	Comments
Thread Plug	(6 to 36) mm	2.5 µm	Labmaster, wires, & gage blocks
Pin & Plug Gages	Up to 25 mm Up to 25 mm Up to 100 mm	1.1 µm 2.6 µm 0.32 µm + 0.016 µm/mm	High accuracy micrometer Micrometer, Labmaster
Rings (Inside Diameter)	(6 to 120) mm	1.0 µm + 0.014 µm/mm	Labmaster, gage blocks, & master ring
Digital Protractors	(0 to 90)°	0.053°	Sine bar and gauge blocks
Calipers ³ – R = 0.01 mm	Up to 300 mm (>300 to 1525) mm	1.5 µm + 0.034 µm/mm 15 µm + 0.0087 µm/mm	Gage blocks
Depth Gage ³ – R = 0.01 mm	Up to 300 mm	11 µm	Gage blocks

Parameter/Equipment	Range	CMC ² (±)	Comments
Depth Micrometers ³ – R = 0.01 mm	Up to 300 mm	12 µm	Gage blocks
OD Micrometers ³ – R = 0.001 mm R = 0.01 mm	Up to 250 mm Up to 250 mm	1.1 µm + 0.011 µm/mm 6.8 µm + 0.00058 µm/mm	Gage blocks
Height Gages ³ – R = 0.01 mm	Up to 1016 mm	34 µm + 0.058 µm/mm	Gage blocks
Indicators (Dial, Digital, & Test) ³ – R = 0.001 mm R = 0.01 mm	Up to 51 mm Up to 51 mm	2.0 µm 4.6 µm	Indicator calibrator

II. Electrical – DC/Low Frequency

Parameter/Equipment	Range	CMC ² (±)	Comments
Electrical Simulation of Thermocouple Indicating Devices – Generate ³ R = 0.1 °C R = 1 °C R = 0.1 °F R = 1 °F	 (-200 to 0) °C (0 to 1370) °C (-200 to 0) °C (0 to 1370) °C (-328 to 32) °F (32 to 2498) °F (-328 to 32) °F (32 to 2498) °F	 1.4 °C 1.0 °C 1.7 °C 1.3 °C 2.6 °F 1.8 °F 2.7 °F 1.9 °F	 Fluke 724 temperature calibrator w/ Type K TC

III. Fluid Quantities

Parameter/Equipment	Range	CMC ² (±)	Comments
Viscosity – Dip Cups Ford Std Cup #4 Ford Std Cup #5	(100 to 1000) mm ² /s (100 to 1000) mm ² /s	4.7 % of rdg 5.2 % of rdg	Standard oils, thermometer, & stopwatch

IV. Mechanical

Parameter/Equipment	Range	CMC ² (±)	Comments
Scales ³ – $R = 0.01$ lb $R = 0.1$ lb $R = 1$ lb	(0.5 to 250) lb (0.5 to 250) lb (0.5 to 250) lb	0.003 % rdg + 0.0081 lb 0.00042 % rdg + 0.082 lb 0.82 lb	Handbook 44 w/ Class 6 weights
Balances ³ – $R = 0.0001$ g $R = 0.001$ g $R = 0.01$ g $R = 0.1$ g $R = 1$ g	Up to 200 g Up to 500 g Up to 1 kg Up to 2 kg Up to 6 kg	0.012 mg + 0.0024 mg/g 1.2 mg + 0.0012 mg/g 8.7 mg 82 mg 1.2 g	Handbook 44 w/ Class 1 weights
Force Gauges ³ – $R = 1$ lb	Up to 150 lbf	0.66 lbf	Class 6 weights and hanger
Torque Analyzers ³ $R = 0.01$ Nm	(1.4 to 200) Nm	0.28 Nm	Arm & weights
Pressure ³ – Transducers & Pressure Gauges $R = 0.5$ psig	(0 to 100) psig	0.32 psig	Fluke 718

Parameter/Equipment	Range	CMC ² (±)	Comments
Torque Wrenches & Drivers –			
Click Type	(0.45 to 5.65) Nm (3.39 to 45.2) Nm (9 to 113) Nm (27 to 339) Nm (136 to 420) Nm	0.035 Nm 0.18 Nm 0.43 Nm 1.6 Nm 1.8 Nm	CDI manual torque system
Dial Type	(0.45 to 5.65) Nm (3.39 to 45.2) Nm (9 to 113) Nm (27 to 339) Nm (136 to 420) Nm	0.043 Nm 0.24 Nm 0.43 Nm 1.4 Nm 1.7 Nm	

V. Time and Frequency

Parameter/Equipment	Range	CMC ² (±)	Comments
Stopwatch	(1 to 180) min	0.20 s	WWV

¹ This laboratory offers commercial calibration service.

² Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

³ R refers to the resolution of the unit under test.



Accredited Laboratory

A2LA has accredited

ADVANCED TECHNOLOGY SERVICES - HMA

Lincoln, AL

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets any additional program requirements in the field of calibration. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009).



Presented this 19th day of March 2015.

A handwritten signature in blue ink, appearing to read 'J. C. Bunt'.

Senior Director of Quality and Communication
For the Accreditation Council
Certificate Number 1592.04
Valid to March 31, 2017

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.